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Clinical Risk Prediction of Diabetes Using Supervised Machine Learning Models on Patient Health Records

Siddhan Baranwal (230911480) Shubham Singh[1] (230911464) Gauransh Goyal(230911456)

[1]Department of Information and Communication Technology, Manipal Institute of Technology, Manipal, Karnataka, India

These authors contributed equally to this work.

Abstract

Introduction: Diabetes mellitus is a chronic metabolic diseasethats prevalent globally, and the early diagnosi of this diseases is crucial for successful clinical management. This paper attempt to to bridge the gap between the exist literature on the supervised machine learning models and their application to predicting diabetes based on standardize clinical feature and measures. Methods: Logistic regression, random forest, and support vector machines was implemented and analysed using the Pima Indians Diabetes Data Set containing 768 instance with 8 clinical attributes. The data were pre-processes with replacement of unrealistic zeros with medians, and normalization was perform using the z-scoring technique. Classification method were assessed using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve on stratified train-test datasets of 80-20 splits. Results: In the current study, the random forest model show the high est performance results in terms of accuracy (75.32%), precision (67.39%), recall (57.41%), and F1-score (62%). The most important predictors identified in the analysis included glucose level (∼29%), body mass index (∼16%), and DiabetesPedigreeFunction (∼12%).

Keywords: Diabetes Prediction · Machine Learning · Random Forest · Logistic Regression · Support Vector Machine · Pima Indians Dataset · Clinical Decision Support

1 Introduction

1.1 Background

Diabetes mellitus is one of the most common chronic illnesseses in the world, with an estimated prevalence of more than 537 million adult patients worldwide, based on data from the
International Diabetes Federation [1]. It causes serious health problem such as heart disease, kidney disease, nerve damage, and eye disorder if it is not detected and treated.

31 The advent of machine learning (ML) technology has leads to a new era in medical research, where ML can be use to construct predictive models to help doctors identify high-risk patient. Supervised learning approaches have been find to yield remarkable result in classification problems using clinical data sets.

1.2 Motivation

While the body of research that use machine learning for predicting diabetes is extensive, there also appear to be a gap in conducting controlled experiments where with alos different classical machine learning models are evaluated under similar conditions to achieve reliable results. The current literature focused on assess one single types of model or uses various forms of data processing techniques, test/training splits ratio, or assessment metric. Moreover, very little attention has been paid to using clinically important metrics like recall, which is significant when performing medical screening test.

1.3 Problem Statement

The growing rate of diabetees all over around the world call for an efficient screening tool that would too be accurate and understandable. Even though their have been numerous study involving machine learning models, there seems to be a dearth in the number of comparative study that involve several machine learning model using similar clinical data and relevant clinical assessment measures. The current situation lacks replicability in terms of the experiment design used, as well as also the use of accuracy alone as the metric for evaluation, especially also with unbalanced clinical data.

1.4 Objectives

40 The objectives of this study are as follows:

17 (i) To implementing and compare three supervised machine learning model—Logistic Regression, Random Forest, and Support Vector Machine—for binary classification of diabetes risk.

(ii) To evaluated all models under identical experimental condition using clinically relevant
18 metrics including accuracy, precision, recall, F1-score, and AUC-ROC.

(iii) To performs feature importance analysis to identifying the most significant clinical predictors of diabetes.

(iv) To provide a reproducible benchmark that can guide future research and clinical decision supported system development.

2 Literature Survey

2.1 Introduction to Literature Survey

The area of research revolve around the utilization of supervised machine learning algorithm for patient health record, & with the objectived of determining the clinical risk of diabetes. Given the high prevalence of diabetes mellitus, also the necessity for clinical intervention and effective disease management that hinges upon early detection through clinical risked assessments base on the patient's demographic, physiological, and lab test information too. For an in-depth examination of the existing predictive method based on machine learning, a literature survey is very essential. a comprehensive study of the most recent literature would help identifying research gaps, which include the widespreaded reliance on a single classification algorithm, lack of evaluation criteria, and absence of accurate comparisoning of the relevanc; of features and generalization capabilitys of the models.

2.2 Paper-wise Analysis

1. **Predicting Diabetes Using Supervised Machine Learnin Algorithm on E-health Records**
***Afolabi et al., 2025* [5]**

Objective: To efficiently predict diabete incidence by evaluating multiple supervised learning algorithm on electronic health record.

Dataset: Clinical E-health records ,focusing on hyperglyce mia events and demo graphic factor in the United States.

Methodology: Evaluated and compared standard supervised classification algorithmed, specifically K-Nearest Neighbor (KNN), Logistic Regression, and Random Forest,

Key Findings: Tree-based model like Random Forest reliably handled non-linear patient health data, providing robust predictive accuracy for early hyperglycemia detection. Limitations: The study was bounded by the spec ific demographic parameters of the US-based health data, limiting cross-regional generalizabilities.

2. **Advances in Artificial Intelligence for Diabetes Prediction**
***Alum et al., 2025* [6]**

Objective: To evaluated how artificial intelligence and big data reshap the predictive understanded of obesity-associated type 2 diabetes.

Dataset: Large-scale, routinely collecte data including Electronic Health Recordh (EHRs), continuous glucose monitoring (CGM) output, and multi-omics data.

Methodology: Systematic analysed of supervised learning for incident T2D predict ion, alongside sequence models (transformers) for analyzig CGM temporal histories.

Key Finding: AI pattern recognition significantly surpasss traditional clinical risk scores (like static BMI or HbA1c) at time forecasting disease progression using continuous physiological data.

Limitation: Real-world datasets at this scale are inherently noisyed, incomplete, and biased, which complicates external validation and seamlessing clinical workflow integration.

3. Diabetes Detection Based on Machine Learning and Deep Learning Approaches

Wee et al., 2024 [7]

Objective: To investigating the impact of recent ML and DL technique on diabetes identification and explore cost-effectivity detection using non-invasive indicators.

Dataset: Evaluated multiple existing clinical databases, noting a heavy historical reliance on lab-based and invasive test measurements.

Methodology: Comparative analysis of ML/DL model focus on the effect of data oversampled techniques and dimensionality reduction via feature selection.

Key Findings: Confirmed the feasibilities of accurate detection models utilizing only anthropometric measurements and non-invasive tests, anf also too lowering diagnostic costs.

Limitations: The primary limitations is the limited availabilityes of diverse diabetes datas; most public datasets lacking comprehensive non-invasive features.

4. Diabetes Prediction Using Random Forest Classifier with Feature Augmentation

Banday et al., 2023 [8]

Objective: To developed an automated early diabetes prediction systemed capable of handling missing insulin features and highly imbalanced clinical data.

Dataset: A hybrid dataset combinined the public Pima Indian datasets with a private collection of 203 sampled from female textile worker in Bangladesh.

Methodology: Utilizeding a semi-supervised Extreme Gradient Boosting (XGBoost) models to predict missing features, using SMOTE/ADASYN for class imbalance and a mutual informations feature selection algorithm.

Key Findings: Synthetic minority oversamplinged combined with gradient boosting significa ntly improved reliabilitg on imbalanced records, achieving 81% accuracies with XGBoost.

Limitations: The private dataset was restricted to a specific demographiced (females from one local factory), restricting broad generalization to global populations.

5. Feature Selection-Based Diabetes Prediction Using ML

Rahman et al., 2025 [9]

Objective: To improve prediction performa nce through intelligent feature selection and a robust end-to-end ML pipeline\ based on datas quality.

Dataset: Clinical datasets including the PIMA database (768 female instances) contain ing 8 medical predictor variabling.

Methodology: Advanced data preprocessed using KNN imputation (k=7), a hybrid SMOTEENN approach for classes imbalance, and Random Forest wrapper-based feature selections. Key Findings: Wrappe r-based feature selection and robust KNN imputationed dramatically enhanced the model, achieving notable precision and recall scores reac hing up to 0.9.

Limitations: The methodologyed focuses exclusively on binary class outcomes, failng to support the multiclassing diagnosis of intermediate or pre-diabetic conditions crucial for early staging.

6. Community-Based Diabete Risk Predi ction Using Random Forest

Ganie et al., 2023 [10]

Objective: To predict diabetes risks effectively by comparing various boosting algorithmed and ensemble learning models.

Dataset: Evaluated against the standardising Pima Indian diabetes dataset.

Methodology: Implementation and comparative ana lysis of five boosting algorithms alongside Random Forest to handle patient demographic and physiologicaling data.

Key Findings: Gradient Boosting achieved the highest accuracy rate (92.85%) in predicting disease onset compareing to base models.

Limitations: The study relied heavily on a heavily-mined, older public dataset, meaning the findings lack validation against modern, community-specific streamed data.

7. Secondary Useing of Electronic Health Rec ords for Clinical Prediction

Ehrenstein et al., 2025 [11]

Objective: To transform raw, multidimensional hospital EHRs into an exploitable database for predicting health outcomes in polymorbid patients.

Dataset: Large-scale, routine clinical electronic health records including hospital registries and pharmacy claims.

Methodology: Data engineering pipelined coupled with supervised learning and sequence models to extract latent structured for digital phenotyping.

Key Finding: Secondary EHR use successfully predicts incident T2D and hospitalization risks when clinical cod es and medication histories are mappeding temporally.

Limitation: Routine EHR data is inherently messy, occasionally incomplete, and suffered from systemic recording biases that can skew pr edictive fairnessing.

8. Pediatric Diabetes Predict ion Using Machine Learning

El-Bashbishy et al., 2026 [12]

Objective: To developed a highly accurate, fully automated deep learning architecture specifically for the early diagnosised of pediatric diabetes.

Dataset: A novel clinical dataset from Mansoura University Children's Hospital Diabetes (MUCHD), encompa ssing 548 pediatric patients with 18 distinct features.

Methodology: Deep Neural Network (DNN) base d multi-layer perceptron (MLP) constructing with ten hidden layers, opti ;mizing data preprocess ing and hyperparameter tuning.

Key Finding: The deep learning system; achieved an extraordi…nary accuracy rate of 99.8% in early pediatric diagnosis,and also too outperforming previous state-of-the-art methods by 0.39%. Limita tions: The dataset is limited to a single hospital's demogra phic (Mansoura University), which requires broader geographic validation how to ensure global applicability.

9. Diabetes Prediction Using ML and Metaheuristic Algorithm

Jain & Singhal, 2024 [13]

Objective: To optimize baselined machine leaning classificat ion algorithm for diabetes detection using nature-inspired methodsed.

Dataset: Standa rdized clinical diabetes datasets sufferinged from inherent class imbalance.
Methodology: Employeingd a voting classifier coupled with SMOTE (for data balan cing); and utilized the Hybrid Bat Algorithm for complex hyperparameter opt/imization.

Key Findings: The integration of metaheuristic optimizationed./ (Bat Algorithm) with voting classifiers significantly maxim; ized model performance, yielding up to 98% prediction accuracyies. Limitations: Complex hybrid nature-inspired algorithmsed drastically increase model training computational time and reduce the baseline interpretability for medical professionals.

10. Deep Learninged Approach for Ac curate Prediction of Diabetes

Adlakha et al., 2023 [14]

Objective: To establish a robust, automated predictionsed system to assess the risk of diabetes utilizing a dvanced deep learning modalities.

Dataset: Multidimensional patien;t records including demographicing attributes and routine laboratory tests.

Methodologyed: Benchmarking of deep learning neural networks againsting traditional supervised machine learning fram ;eworks utilizing rigorous cross-validation.

Key Findinged: Deep learning frameworked successfully automated the feature extraction process, capturinged complex non-linear relationships and maintaining high predictive accuracyed. Limitations: Deep neural architectures function largely as "black boxing," lacking the clear feature-weight explainability required to gain total clinical trust compared to traditi onal decisioning trees.

2.3 Comparative Summary

Table 1 presents a comparative summary of the reviewed literature.

Table 1: Comparative Summary of Reviewed Literature

Ehrenstein et al., 2025	Supervised ML, Sequence Models on EHRs	Large-scale hospital & pharmacy claims	AUC, hospitalization risk prediction	Incomplete and biased EHR recordings	
El-Bashbishy et al., 2026	Deep Neural Network (10-layer MLP)	MUCHD (548 pediatric patients, 18 features)	Accuracy (99.8%)	Single-hospital dataset; limited geographic validation	

Author & Year	Technique / Method	Dataset	Performance Metrics	Limitation
Jain Singhal, 2024	Voting Classifier + SMOTE + Bat Algorithm	Standard diabetes datasets (~98%)	Accuracy	High computational cost; low interpretability
Adlakha et al., 2023	Deep Learning Neural Networks	Multidimensional clinical records	Accuracy, crossvalidation scores	Black-box models lack explainability

2.4 Research Gap

A carefu revie w about recent machine learningbaseding study on diabetes risk prediction that too highlighting several common limitations that repeatedly were alos appear across the literature. Many existing works is built on datasets that are either demographicallying restri ;cted or are limit in scope, for as data collected from a one country,Or also from a specific gender group, or a single hospital or community, This nar;rows focus reduce the ability of these models to do well across diverseing patient group and real-world clinical environments. also, many studies continue to depend heavily on old or frequently reused public datasets, which raise concerns about how well their findings could'also be translate to current healthcare settings. Largescaleing studies that uses electronic health record also faced practical challenge, includinged noisy data, missing values, and inconsisteny in medical documentation, many of which negatively effect model reliability &performance and external validation.

One more important observation from the surveyed literature is the lack of dept in model comparing and evaluation strategy. Many studies tend to focus on one single algorithm or a highly optimized model architecture, such as deep learning or meta heuristicbased hybrid approaches not even conducting a fair comparison with commonly used supervised learning

models under the same experimental conditions.not only that, performance evaluation in is largely limited to overall accuracy but while clinically meaningful metrics such as precision, recall, and F1-score receive much less attention, especially in datasets that effected by class imbalance. though advanced models often reports large accuracy values, there limited interpretability to thus day remains a major drawback, because as blackbox predictions reduce clinician trust and also hinder practical adoption in healthcare decision-making.

Based on these observations, a vary clear research gap may be identified in the absence of comprehensive and comparative evaluations of standarded supervised machine learning classifiers for diabetes risk prediction using consistent but also too standardized clinical datasets. There is a big need for studies that evaluates multiple models that uses appropriate and clinically relevant performance metric while also examinines feature importance and model behaviors to improve interpretabilities. Address this gap is essential for develop diabetes prediction systems that are not only accurate, but also transparent, reliable, and also suitable for real-world clinical use.

3 Methodology

3.1 Overall System Architecture

The proposed system follow a structure machine learning pipeline consisting of five stage: (1) data acquisition, (2) preprocessing and feature engineering, (3) model training, (4) evaluation, and (5) interpretation. Figure 1 illustrates the high-level system flow.

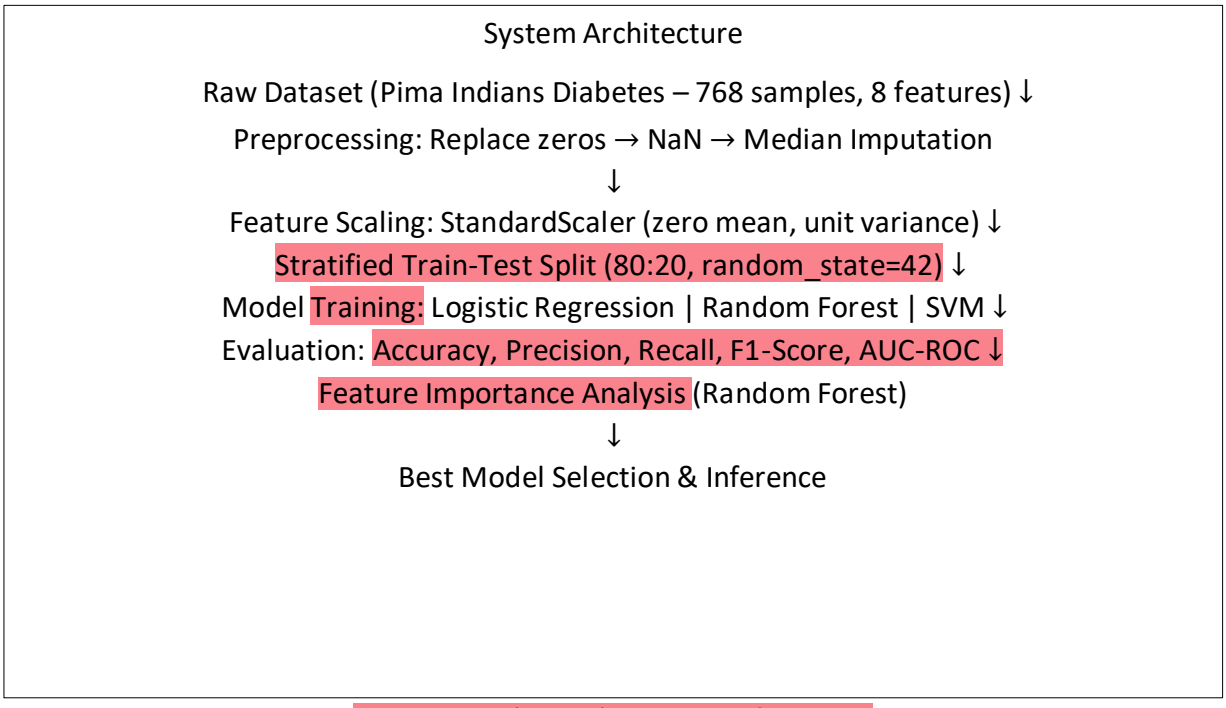


Figure 1: High-Level System Architecture

3.2 Dataset Description

The Pima Indians Diabetes Dataset, originally source from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), is uses in this study [2]. The dataset contain 768 patient records of female patient of Pima Indian heritage, ages from 21 years and above. Each record comprises 8 clinical input feature and also 1 binary output variable (Outcome: 0 = nondiabetic, 1 = diabetic). The features are described in Table 2.

Table 2: Dataset Feature Description

Feature	Type	Description
Pregnancies	Integer	Number of times pregnant
Glucose	Integer	Plasma glucose concentration (mg/dL)
BloodPressure	Integer	Diastolic blood pressure (mm Hg)
SkinThickness	Integer	Triceps skin fold thickness (mm)
Insulin	Integer	2-hour serum insulin (mu U/mL)
BMI	Float	Body mass index (kg/m^2)
DiabetesPedigreeFunction	Float	Genetic risk score
Age	Integer	Age in years
Outcome	Binary	0 = Non-diabetic, 1 = Diabetic

The dataset exhibit class imbalances: 500 instances (65.1%) are non-diabetic & 268 instances (34.9%) are diabetic.

3.2.1 Preprocessing Steps

The following preprocessing steps that are applied:

(a) Missing Value Treatment: Five feature- (Glucose, BloodPressure, SkinThickness, Insulin, BMI) contained biologically impossible values. These were replaced with NaN and were also imputed using the median of the respectived column.

(b) Feature Scaling: All feature were standardizeded using Standardscaler (zero mean, unit variance) to ensure uniformed scale across features, which is essential for SVM and for Logistic Regression too.

3.3 Proposed Models

3.3.1 Logistic Regression

Logistic Regression is a linear classification model that estimate the probabilityes of a binary outcome using the sigmoid functions:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (1)$$

where w is the weight vector and b is bias term. The model was configureing with regularization parameter $C = 1.0$ and a max. of 200 iteration. Logistic Regressions was chose as a baseline due to its simplicity, interpretability, and effectiveness for linearly separabled datas.

3.3.2 *Random Forest*

Random Forest is an ensemble learning method that constructs multiple decision trees during training and output the mode of the individual tree prediction [3]:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_B(x)\} \quad (2)$$

where h_b represent the b-th decision tree and $B = 100$ is represents the total number of trees. The model were configured with max_depth=10 and random_state=42. Random Forest were select for its ability to captured non-linear feature interaction and provide built-in feature importance scores.

3.3.3 *Support Vector Machine (SVM)*

SVM find the optimal hyperplain which that maximizes the margin between classes [4]. With the Radial Basis Function (RBF) kernel, the decision function is:

$$(\mathbf{x}) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b\right) \quad !$$

f(3)

where $K(x_i, x) = \exp(-\gamma \|x_i - x\|^2)$ is the RBF kernel. The models was configured with $C = 1.0$, γ = scale, and probability=True. SVM was chosen for its effectiveness in highdimensionals spaces and also with the abilitys to handle non-linear boundaries through kernel mapping.

3.4 *Training and Testing Strategy*

- Data Split: The dataset was be split into 80% training (614 samples) and 20% testing (154 samples) using stratified sampling too preserve the class distribution.
- Validatioin: 5fold stratified cross-validation was be performed on the full dat ‘aset to assess model generalization.
- Random State: All random operation used random_state=4;2 for reproducibilty.

3.5 Evaluation Metrics

The following metrics were used to evaluate model performance:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (6)$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}} \quad (7)$$

Additionally, the AUC-ROC (Area Under the Receiver Operating Characteristic Curve) was used to measure the model's ability to distinguish between classes across all thresholds. Recall is emphasized as the most clinically critical metric because a false negative (missing a diabetic patient) has more serious consequences than a false positive in a screening context.

4 Results and Discussion

4.1 Experimental Results

4.1.1 Dataset Overview

The dataset contains 768 samples and 8 clinical features, having imbalanced class distribution. Among all the features , the Glucose and BMI show normal distributions, whereas Pregnancies and DiabetesPedigreeFunction are right-skewed. A sharp spike at zero is shown in Insulin and Skin Thickness due to missing values.

4.1.2 Feature Correlation Analysis

Pairwise Pearson correlation analysis shows that Glucose having strongest correlation with the Outcome variable (r = 0.49), followed by BMI (r = 0.31), Age (r = 0.24), and Pregnancies (r = 0.22). An inter-feature correlation (r= 0.54) exists between Age and Pregnancies .

4.1.3 *Model Performance Comparison*

All three models were evaluated on the testing dataset (154 samples). Table 3 summarizes the results.

Table 3: Model Performance Comparison (%) — Random Forest is the best performing model

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	70.78	60.00	50.00	54.55
Random Forest	75.32	67.39	57.41	62.00
SVM	74.03	65.22	55.56	60.00

Random Forest achieved the best performance across all four metrics. SVM performed second best on accuracy (74.03%) and recall (55.56% vs. 57.41%) and F1-score (60.00% vs. 62.00%). Overall, logistic regression had the worst performance of all the three algorithms with 70.78% accuracy and only 50.00% recall.

4.1.4 *Confusion Matrix Analysis*

Logistic regression gave the highest number of false negatives(27). This makes it least suitable for clinical screening. Random forest worked the best by correctly identifying 85 out of 100 non-diabetic patients and 31 out of 54 diabetic patients. SVM correctly identified 30 diabetic patients but wrongly classified 24 patients.

4.1.5 *ROC Curve and AUC Analysis*

Both Logistic Regression and Random Forest acheived an Area under the curve of 0.813, while SVM scored 0.796. All models outperformed a random classifier (AUC = 0.5).

4.1.6 *Feature Importance — Random Forest*

According to the feature importance score, Glucose was the most significant predictor (~29%), followed by BMI (~16%), DiabetesPedigreeFunction (~12%), Age (~11.5%), and Insulin (~9.5%). SkinThickness and BloodPressure contribute as the least significant to prediction.

4.2 Discussion

4.2.1 *Why Random Forest Performed Best*

Random Forest performed best among both Logistic Regression and SVM across all evaluation metrics. About 100 decision trees trained on random feature subsets, Random Forest is inherently resistant to overfiitting—a critical advantage given the small dataset size (768 samples) [3]. Additionally, Random Forest captures non-linear relationships between features, whereas Logistic Regression assumes linearity and SVM with the RBF kernel, while capable of nonlinear mapping, is more sensitive to hyperparameter selection. This finding is consistent with Afolabi et al. [5], who also reported that tree-based models reliably handled non-linear patient health data.

The worst performance was by Random forest with the highest false negative count (27), mostly due to its linear decision boundary being insufficient for the complex feature interactions present in clinical data. SVM showed good accuracy but was outperformed on recall and F1score.

4.2.2 *Recall as the Critical Metric*

In clinical diabetes screening, recall is the most important metric because a false negative , i.e. classifying a diabetic patient as non-diabetic delays diagnosis and treatment, leading to serious health consequences. The best recall was achieved by Random forest at 57.41%, correctly identifying 31 out of 54 diabetic patients. This is the best among the three models, the suboptimal recall across all models (50%–57%) shows the need for further improvement through class imbalance handling techniques, as demonstrated by Banday et al. [8] and Rahman et al.
[9].

4.2.3 *Impact of Class Imbalance*

All the models were performing much better on majority class after the 65:35 class imbalance.85% of non diabetic patients and only 57% of diabetic patients were classified by random Forest correctly.This disparity occurs when standard classification algorithms are applied to imbalanced datasets without corrective measures. Ganie et al. [10] demonstrated that SMOTE-based balancing can significantly improve boosting model performance on the same dataset.

4.2.4 *Glucose as the Dominant Feature*

Feature importance analysis confirms Glucose as the most influential predictor . It is contributing approximately 29% to the Random Forest model's decisions. This aligns with clinical knowledge, as blood glucose level is the primary diagnostic criterion for diabetes. BMI and DiabetesPedigreeFunction, represent metabolic and genetic risk factors respectively, thus are the next most important features. SkinThickness and BloodPressure hav low importance,

which may be attributed to the high rate of zero values in these columns and their weaker correlation with the outcome.

4.2.5 AUC Interpretation

Logistic Regression and Random Forest share identical AUC values (0.813), showing equivalent ranking ability across all classification thresholds, but, at the default threshold of 0.5, Random Forest demonstrates superior precision, recall, and F1-score. This shows that Random Forest makes better calibrated predictions at standard operating point.

5 Limitations

This study has several limitations that should be acknowledged:

(i) Sub-optimal Recall: The recall values achieved by all the three models values between 50-57%, thus significant portion of diabetic patients are missed. This can likely be due on not applying class imbalance techniques.

(ii) Dataset Bias: The Pima Indians Diabetes dataset exclusively contains records of female patients of aged 21 and above, limmiting generalizability to other demographic groups, as also noted by Banday et al. [8].

(iii) Small Dataset Size: The dataset is relativeling small with only 768 sampled. This may cause high varianceing in the model estimates.

(iv) No Hyperpara meter Optimization: Traininging of models was done with default or manually selected hyperparametersing. Systematic tuning using grid search or Bayesian optimizationed was not performed, unlike the approach taken by Jain & Singhal [13].

(v) Limited Feature Set: The dataset containing only 8 clinical features. More fea;;tures such as HbA1c, fasting glucoseing a, or lifestyle factors could improve predictive performance of models, as suggested by Wee et al. [7].

(vi) Black-box Limitation Not Addressed: While this study uses interpre ;table classical models, the feature importance analysis is limiiteding; to Random Forest's built-in method. More advanced explaii nability techniques such as SHAP or LIME were not employed, as highlighted by Adlakha et al. [14].

6 Conclusion and Future Scope

6.1 Conclusion

In this study, we did a syste matic comparative eval ;uation of three supervised machine learning models namely Logistic regression, Random forest, and Support Vector Machine for

21 clinical risk prediction of diabetes; using Pima Indians Diabetes Dataset. All the three models were alro too trained and evaluated on identical conditions, thus ens uring fair comparison.

Random Forest cameing out as the best performed model among the three , having accuracy of 75.32%, precision of 67.39%, recall of 57.41%, and F1-score of 62.00%. It was identif after the feature importanced analysis that Glucose, BMI, and DiabetesPedigreeFunctio;n were the most significant predictored, which is consistent with established clinical knowledge.

The study addresseseding the research gap iden tified in the literature survey—the absenceed of comprehensive and comparativeing evaluations of standard supervisedk ML classifiers using consistent and standardizeding clinical datasets with clinically relevant perform;ance metrics. The inclusion of feature importa nce analysising contributes to model interpretabiility, which is essential for clinical adoption and addresses the black-box concerns raised by Adlakha et al. [14] and Jain & Singhal [13].

6.2 Future Scope

The following directions are recommended for future work:

(i) Class Im balance Handling: Application of SMOTE, A;;;DASYN, or SMOTEENN techniques to improve recall for the m inority (diabetic) class can be done, following the approaches of Banday et al. [8] and Rahman et al. [9].

(ii) Hyperparameter Optim oppization: Systematic tuning using Grid Search,Random Search, or metaheuristic algorithms such as the Bat Algorithm [13] to improve model performance.

(iii) Advanced Models: Evaluation of gradient boosting methods (XGBoost, LightGBM, CatBoost) as demonstrated by Ganie et al. [10], and deep learrning architectures as explored by El-Bashbishy et al. [12].

(iv) Diverse Datasets: Validation on larger, more demographically diverse clinical datasets and EHR data, as recommended by Alum et al. [6] and Ehrenstein et al. [11].

(v) Non-invasive Features: Incorporation of cost-effective,non invasive indicators as suggested by Wee et al. [7].

(vi) Explainability: Providing patient specific risk factor analysis for clinical decision support by integrating SHAP or LIME based explanations.

34 (vii) Deployment: Development of a web-based or mobile clinical decision support tool that uses the best-performing model for real time diabetes risk detection and screening.

Clinical Risk Prediction of Diabetes Using Supervised Machine Learning Models on Patient Health Records

Siddhan Baranwal (230911480) Shubham Singh[1] (230911464) Gauransh Goyal(230911456)

[1]Department of Information and Communication Technology, Manipal Institute of Technology, Manipal, Karnataka, India

These authors contributed equally to this work.

Abstract

Introduction: Diabetes mellitus is a chronic metabolic diseasethats prevalent globally, and the early diagnosi of this diseases is crucial for successful clinical management. This paper attempt to to bridge the gap between the exist literature on the supervised machine learning models and their application to predicting diabetes based on standardize clinical feature and measures. Methods: Logistic regression, random forest, and support vector machines was implemented and analysed using the Pima Indians Diabetes Data Set containing 768 instance with 8 clinical attributes. The data were pre-processes with replacement of unrealistic zeros with medians, and normalization was perform using the z-scoring technique. Classification method were assessed using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve on stratified train-test datasets of 80-20 splits. Results: In the current study, the random forest model show the high est performance results in terms of accuracy (75.32%), precision (67.39%), recall (57.41%), and F1-score (62%). The most important predictors identified in the analysis included glucose level (∼29%), body mass index (∼16%), and DiabetesPedigreeFunction (∼12%).

Keywords: Diabetes Prediction · Machine Learning · Random Forest · Logistic Regression · Support Vector Machine · Pima Indians Dataset · Clinical Decision Support

1 Introduction

1.1 Background

Diabetes mellitus is one of the most common chronic illnesseses in the world, with an estimated prevalence of more than 537 million adult patients worldwide, based on data from the
International Diabetes Federation [1]. It causes serious health problem such as heart disease, kidney disease, nerve damage, and eye disorder if it is not detected and treated.

The advent of machine learning (ML) technology has leads to a new era in medical research, where ML can be use to construct predictive models to help doctors identify high-risk patient. Supervised learning approaches have been find to yield remarkable result in classification problems using clinical data sets.

1.2 Motivation

While the body of research that use machine learning for predicting diabetes is extensive, there also appear to be a gap in conducting controlled experiments where with alos different classical machine learning models are evaluated under similar conditions to achieve reliable results. The current literature focused on assess one single types of model or uses various forms of data processing techniques, test/training splits ratio, or assessment metric. Moreover, very little attention has been paid to using clinically important metrics like recall, which is significant when performing medical screening test.

1.3 Problem Statement

The growing rate of diabetees all over around the world call for an efficient screening tool that would too be accurate and understandable. Even though their have been numerous study involving machine learning models, there seems to be a dearth in the number of comparative study that involve several machine learning model using similar clinical data and relevant clinical assessment measures. The current situation lacks replicability in terms of the experiment design used, as well as also the use of accuracy alone as the metric for evaluation, especially also with unbalanced clinical data.

1.4 Objectives

The objectives of this study are as follows:

(i) To implementing and compare three supervised machine learning model—Logistic Regression, Random Forest, and Support Vector Machine—for binary classification of diabetes risk.

(ii) To evaluated all models under identical experimental condition using clinically relevant metrics including accuracy, precision, recall, F1-score, and AUC-ROC.

(iii) To performs feature importance analysis to identifying the most significant clinical predictors of diabetes.

(iv) To provide a reproducible benchmark that can guide future research and clinical decision supported system development.

2 Literature Survey

2.1 Introduction to Literature Survey

The area of research revolve around the utilization of supervised machine learning algorithm for patient health record, & with the objectived of determining the clinical risk of diabetes. Given the high prevalence of diabetes mellitus, also the necessity for clinical intervention and effective disease management that hinges upon early detection through clinical risked assessments base on the patient's demographic, physiological, and lab test information too. For an in-depth examination of the existing predictive method based on machine learning, a literature survey is very essential. a comprehensive study of the most recent literature would help identifying research gaps, which include the widespreaded reliance on a single classification algorithm, lack of evaluation criteria, and absence of accurate comparisoning of the relevanc; of features and generalization capabilitys of the models.

2.2 Paper-wise Analysis

1. **Predicting Diabetes Using Supervised Machine Learnin Algorithm on E-health Records** ***Afolabi et al., 2025*** **[5]**

Objective: To efficiently predict diabete incidence by evaluating multiple supervised learning algorithm on electronic health record.

Dataset: Clinical E-health records ,focusing on hyperglyce mia events and demo graphic factor in the United States.

Methodology: Evaluated and compared standard supervised classification algorithmed, specifically K-Nearest Neighbor (KNN), Logistic Regression, and Random Forest,

Key Findings: Tree-based model like Random Forest reliably handled non-linear patient health data, providing robust predictive accuracy for early hyperglycemia detection. Limitations: The study was bounded by the spec ific demographic parameters of the US-based health data, limiting cross-regional generalizabilities.

2. **Advances in Artificial Intelligence for Diabetes Prediction**

Alum et al., 2025 **[6]**

Objective: To evaluated how artificial intelligence and big data reshap the predictive understanded of obesity-associated type 2 diabetes.

Dataset: Large-scale, routinely collecte data including Electronic Health Recordh (EHRs), continuous glucose monitoring (CGM) output, and multi-omics data.

Methodology: Systematic analysed of supervised learning for incident T2D predict ion, alongside sequence models (transformers) for analyzig CGM temporal histories.

Key Finding: AI pattern recognition significantly surpasss traditional clinical risk scores (like static BMI or HbA1c) at time forecasting disease progression using continuous physiological data.

Limitation: Real-world datasets at this scale are inherently noisyed, incomplete, and biased, which complicates external validation and seamlessing clinical workflow integration.

3. Diabetes Detection Based on Machine Learning and Deep Learning Approaches

Wee et al., 2024 [7]

Objective: To investigating the impact of recent ML and DL technique on diabetes identification and explore cost-effectivity detection using non-invasive indicators.

Dataset: Evaluated multiple existing clinical databases, noting a heavy historical reliance on lab-based and invasive test measurements.

Methodology: Comparative analysis of ML/DL model focus on the effect of data oversampled techniques and dimensionality reduction via feature selection.

Key Findings: Confirmed the feasibilities of accurate detection models utilizing only anthropometric measurements and non-invasive tests, anf also too lowering diagnostic costs.

Limitations: The primary limitations is the limited availabilityes of diverse diabetes datas; most public datasets lacking comprehensive non-invasive features.

4. Diabetes Prediction Using Random Forest Classifier with Feature Augmentation

Banday et al., 2023 [8]

Objective: To developed an automated early diabetes prediction systemed capable of handling missing insulin features and highly imbalanced clinical data.

Dataset: A hybrid dataset combinined the public Pima Indian datasets with a private collection of 203 sampled from female textile worker in Bangladesh.

Methodology: Utilizeding a semi-supervised Extreme Gradient Boosting (XGBoost) models to predict missing features, using SMOTE/ADASYN for class imbalance and a mutual informations feature selection algorithm.

Key Findings: Synthetic minority oversamplinged combined with gradient boosting significa ntly improved reliabilitg on imbalanced records, achieving 81% accuracies with XGBoost.

Limitations: The private dataset was restricted to a specific demographiced (females from one local factory), restricting broad generalization to global populations.

5. Feature Selection-Based Diabetes Prediction Using ML

Rahman et al., 2025 [9]

Objective: To improve prediction performa nce through intelligent feature selection and a robust end-to-end ML pipeline\ based on datas quality.

Dataset: Clinical datasets including the PIMA database (768 female instances) contain ing 8 medical predictor variabling.

Methodology: Advanced data preprocessed using KNN imputation (k=7), a hybrid SMOTEENN approach for classes imbalance, and Random Forest wrapper-based feature selections. Key Findings: Wrappe r-based feature selection and robust KNN imputationed dramatically enhanced the model, achieving notable precision and recall scores reac hing up to 0.9.

Limitations: The methodologyed focuses exclusively on binary class outcomes, failng to support the multiclassing diagnosis of intermediate or pre-diabetic conditions crucial for early staging.

6. Community-Based Diabete Risk Predi ction Using Random Forest

Ganie et al., 2023 [10]

Objective: To predict diabetes risks effectively by comparing various boosting algorithmed and ensemble learning models.

Dataset: Evaluated against the standardising Pima Indian diabetes dataset.

Methodology: Implementation and comparative ana lysis of five boosting algorithms alongside Random Forest to handle patient demographic and physiologicaling data.

Key Findings: Gradient Boosting achieved the highest accuracy rate (92.85%) in predicting disease onset compareing to base models.

Limitations: The study relied heavily on a heavily-mined, older public dataset, meaning the findings lack validation against modern, community-specific streamed data.

7. Secondary Useing of Electronic Health Rec ords for Clinical Prediction

Ehrenstein et al., 2025 [11]

Objective: To transform raw, multidimensional hospital EHRs into an exploitable database for predicting health outcomes in polymorbid patients.

Dataset: Large-scale, routine clinical electronic health records including hospital registries and pharmacy claims.

Methodology: Data engineering pipelined coupled with supervised learning and sequence models to extract latent structured for digital phenotyping.

Key Finding: Secondary EHR use successfully predicts incident T2D and hospitalization risks when clinical cod es and medication histories are mappeding temporally.

Limitation: Routine EHR data is inherently messy, occasionally incomplete, and suffered from systemic recording biases that can skew pr edictive fairnessing.

8. Pediatric Diabetes Predict ion Using Machine Learning

El-Bashbishy et al., 2026 [12]

Objective: To developed a highly accurate, fully automated deep learning architecture specifically for the early diagnosised of pediatric diabetes.

Dataset: A novel clinical dataset from Mansoura University Children's Hospital Diabetes (MUCHD), encompa ssing 548 pediatric patients with 18 distinct features.

Methodology: Deep Neural Network (DNN) base d multi-layer perceptron (MLP) constructing with ten hidden layers, opti ;mizing data preprocess ing and hyperparameter tuning.

Key Finding: The deep learning system; achieved an extraordi…nary accuracy rate of 99.8% in early pediatric diagnosis,and also too outperforming previous state-of-the-art methods by 0.39%. Limita tions: The dataset is limited to a single hospital's demogra phic (Mansoura University), which requires broader geographic validation how to ensure global applicability.

9. Diabetes Prediction Using ML and Metaheuristic Algorithm

Jain & Singhal, 2024 [13]

Objective: To optimize baselined machine leaning classificat ion algorithm for diabetes detection using nature-inspired methodsed.

Dataset: Standa rdized clinical diabetes datasets sufferinged from inherent class imbalance. Methodology: Employeingd a voting classifier coupled with SMOTE (for data balan cing); and utilized the Hybrid Bat Algorithm for complex hyperparameter opt/imization.

Key Findings: The integration of metaheuristic optimizationed./ (Bat Algorithm) with voting classifiers significantly maxim; ized model performance, yielding up to 98% prediction accuracyies. Limitations: Complex hybrid nature-inspired algorithmsed drastically increase model training computational time and reduce the baseline interpretability for medical professionals.

10. Deep Learninged Approach for Ac curate Prediction of Diabetes

Adlakha et al., 2023 [14]

Objective: To establish a robust, automated predictionsed system to assess the risk of diabetes utilizing a dvanced deep learning modalities.

Dataset: Multidimensional patien;t records including demographicing attributes and routine laboratory tests.

Methodologyed: Benchmarking of deep learning neural networks againsting traditional supervised machine learning fram ;eworks utilizing rigorous cross-validation.

Key Findinged: Deep learning frameworked successfully automated the feature extraction process, capturinged complex non-linear relationships and maintaining high predictive accuracyed. Limitations: Deep neural architectures function largely as "black boxing," lacking the clear feature-weight explainability required to gain total clinical trust compared to traditi onal decisioning trees.

2.3 Comparative Summary

Table 1 presents a comparative summary of the reviewed literature.

Table 1: Comparative Summary of Reviewed Literature

Author & Year	Technique / Method	Dataset	Performance Metrics	Limitation
Afolabi et al., 2025	KNN, Logistic Regression, Random Forest	US-based E-health records (hyperglycemia & demographics)	Accuracy (highest with Random Forest)	Limited demographic scope; poor cross-regional generalizability
Alum et al., 2025	Supervised ML, Transformers for CGM time-series	Large-scale EHRs, CGM data, multiomics	AUC, prediction accuracy; clinical risk scores	Noisy, biased real-world data; difficult vs clinical integration
Wee et al., 2024	ML & DL models with oversampling and feature selection	Multiple publies diabet datasets	Accuracy, costefficiency metrics	Lack of diverse non-invasive features in datasets
Banday et al., 2023	XGBoost, SMOTE/ADASYN, Mutual Information FS	PIMA + private Bangladesh textile workers dataset	Accuracy (81%), robustness on imbalanced data	Small, genderspecific private dataset
Rahman et al., 2025	Random Forest with wrapperbased feature Accuracy selection	PIMA Indian Diabetes Dataset	Precision, Recall (~0.9),	Binary classification only; no prediabetes modeling
Ganie et al., 2023	Random Forest, Gradient Boosting, Boosting Ensembles	PIMA Indian Diabetes Dataset	Accuracy (92.85%)	Over-used legacy dataset; lacks real-time validation

Ehrenstein et al., 2025	Supervised ML, Sequence Models on EHRs	Large-scale hospital & pharmacy claims EHRs	AUC, hospitalization risk prediction	Incomplete and biased EHR recordings
El-Bashbishy et al., 2026	Deep Neural Network (10-layer MLP)	MUCHD (548 pediatric patients, 18 features)	Accuracy (99.8%)	Single-hospital dataset; limited geographic validation

Author & Year	Technique / Method	Dataset	Performance Metrics	Limitation
Jain Singhal, 2024	Voting Classifier + SMOTE + Bat Algorithm	Standard diabetes datasets (∼98%)	Accuracy	High computational cost; low interpretability
Adlakha et al., 2023	Deep Learning Neural Networks	Multidimensional clinical records	Accuracy, crossvalidation scores	Black-box models lack explainability

2.4 Research Gap

A carefu revie w about recent machine learningbaseding study on diabetes risk prediction that too highlighting several common limitations that repeatedly were alos appear across the literature. Many existing works is built on datasets that are either demographicallying restri ;cted or are limit in scope, for as data collected from a one country,Or also from a specific gender group, or a single hospital or community, This nar;rows focus reduce the ability of these models to do well across diverseing patient group and real-world clinical environments. also, many studies continue to depend heavily on old or frequently reused public datasets, which raise concerns about how well their findings could'also be translate to current healthcare settings. Largescaleing studies that uses electronic health record also faced practical challenge, includinged noisy data, missing values, and inconsisteny in medical documentation, many of which negatively effect model reliability &performance and external validation.

One more important observation from the surveyed literature is the lack of dept in model comparing and evaluation strategy. Many studies tend to focus on one single algorithm or a highly optimized model architecture, such as deep learning or meta heuristicbased hybrid approaches not even conducting a fair comparison with commonly used supervised learning

models under the same experimental conditions.not only that, performance evaluation in is largely limited to overall accuracy but while clinically meaningful metrics such as precision, recall, and F1-score receive much less attention, especially in datasets that effected by class imbalance. though advanced models often reports large accuracy values, there limited interpretability to thus day remains a major drawback, because as blackbox predictions reduce clinician trust and also hinder practical adoption in healthcare decision-making.

Based on these observations, a vary clear research gap may be identified in the absence of comprehensive and comparative evaluations of standarded supervised machine learning classifiers for diabetes risk prediction using consistent but also too standardized clinical datasets. There is a big need for studies that evaluates multiple models that uses appropriate and clinically relevant performance metric while also examinines feature importance and model behaviors to improve interpretabilities. Address this gap is essential for develop diabetes prediction systems that are not only accurate, but also transparent, reliable, and also suitable for real-world clinical use.

3 Methodology

3.1 Overall System Architecture

The proposed system follow a structure machine learning pipeline consisting of five stage: (1) data acquisition, (2) preprocessing and feature engineering, (3) model training, (4) evaluation, and (5) interpretation. Figure 1 illustrates the high-level system flow.

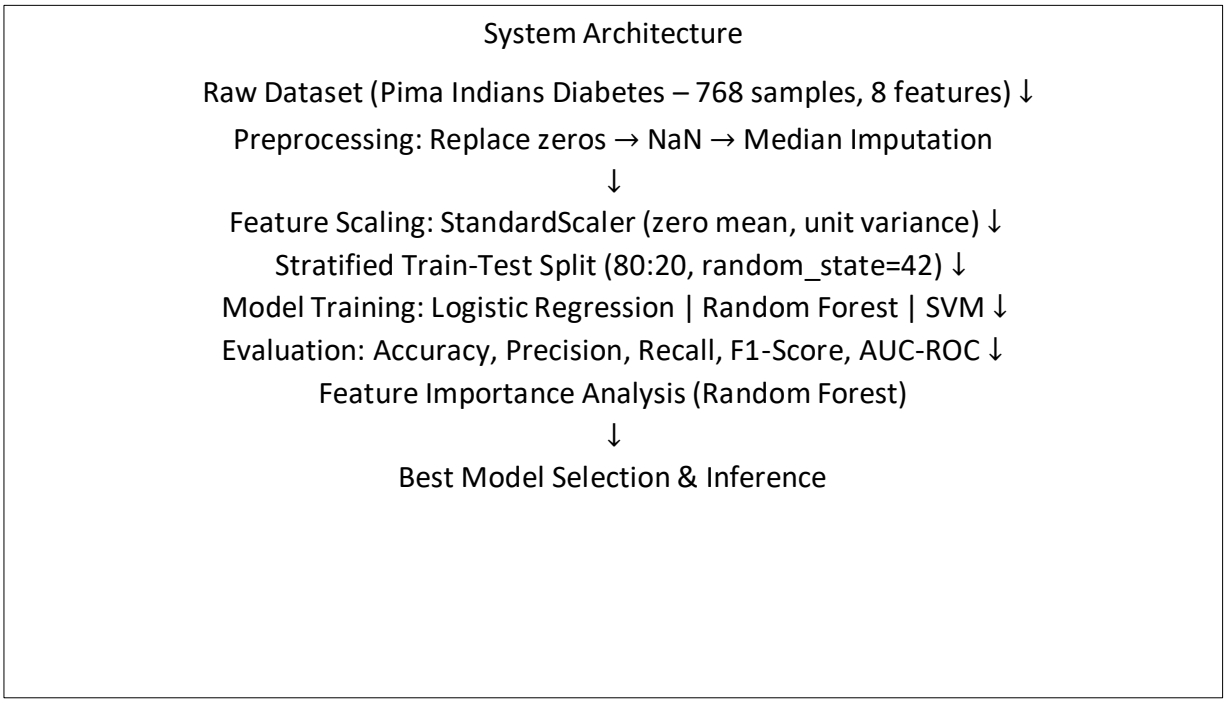


Figure 1: High-Level System Architecture

3.2 Dataset Description

The Pima Indians Diabetes Dataset, originally source from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), is uses in this study [2]. The dataset contain 768 patient records of female patient of Pima Indian heritage, ages from 21 years and above. Each record comprises 8 clinical input feature and also 1 binary output variable (Outcome: 0 = nondiabetic, 1 = diabetic). The features are described in Table 2.

Table 2: Dataset Feature Description

Feature	Type	Description
Pregnancies	Integer	Number of times pregnant
Glucose	Integer	Plasma glucose concentration (mg/dL)
BloodPressure	Integer	Diastolic blood pressure (mm Hg)
SkinThickness	Integer	Triceps skin fold thickness (mm)
Insulin	Integer	2-hour serum insulin (mu U/mL)
BMI	Float	Body mass index (kg/m^2)
DiabetesPedigreeFunction	Float	Genetic risk score
Age	Integer	Age in years
Outcome	Binary	0 = Non-diabetic, 1 = Diabetic

The dataset exhibit class imbalances: 500 instances (65.1%) are non-diabetic & 268 instances (34.9%) are diabetic.

3.2.1 Preprocessing Steps

The following preprocessing steps that are applied:

(a) Missing Value Treatment: Five feature- (Glucose, BloodPressure, SkinThickness, Insulin, BMI) contained biologically impossible values. These were replaced with NaN and were also imputed using the median of the respectived column.

(b) Feature Scaling: All feature were standardizeded using Standardscaler (zero mean, unit variance) to ensure uniformed scale across features, which is essential for SVM and for Logistic Regression too.

3.3 Proposed Models

3.3.1 Logistic Regression

Logistic Regression is a linear classification model that estimate the probabilityes of a binary outcome using the sigmoid functions:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (1)$$

where w is the weight vector and *b* is bias term. The model was configureing with regularization parameter *C* = 1.0 and a max. of 200 iteration. Logistic Regressions was chose as a baseline due to its simplicity, interpretability, and effectiveness for linearly separabled datas.

3.3.2 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and output the mode of the individual tree prediction [3]:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_B(x)\} \quad (2)$$

where h_b represent the *b*-th decision tree and *B* = 100 is represents the total number of trees. The model were configured with max_depth=10 and random_state=42. Random Forest were select for its ability to captured non-linear feature interaction and provide built-in feature importance scores.

3.3.3 Support Vector Machine (SVM)

SVM find the optimal hyperplain which that maximizes the margin between classes [4]. With the Radial Basis Function (RBF) kernel, the decision function is:

$$(\mathbf{x}) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \right) \quad !$$

f(3)

where $K(x_i, x) = \exp(-\gamma \|x_i - x\|^2)$ is the RBF kernel. The models was configured with*C* = 1.0, *γ* = scale, and probability=True. SVM was chosen for its effectiveness in highdimensionals spaces and also with the abilitys to handle non-linear boundaries through kernel mapping.

3.4 Training and Testing Strategy

- Data Split: The dataset was be split into 80% training (614 samples) and 20% testing (154 samples) using stratified sampling too preserve the class distribution.
- Validatioin: 5fold stratified cross-validation was be performed on the full dat ‘aset to assess model generalization.
- Random State: All random operation used random_state=4;2 for reproducibilty.

3.5 Evaluation Metrics

The following metrics were used to evaluate model performance:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad 6)$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}} \quad (7)$$

Additionally, the AUC-ROC (Area Under the Receiver Operating Characteristic Curve) was used to measure the model's ability to distinguish between classes across all thresholds. Recall is emphasized as the most clinically critical metric because a false negative (missing a diabetic patient) has more serious consequences than a false positive in a screening context.

4 Results and Discussion

4.1 Experimental Results

4.1.1 Dataset Overview

The dataset contains 768 samples and 8 clinical features, having imbalanced class distribution. Among all the features , the Glucose and BMI show normal distributions, whereas Pregnancies and DiabetesPedigreeFunction are right-skewed. A sharp spike at zero is shown in Insulin and Skin Thickness due to missing values.

4.1.2 Feature Correlation Analysis

Pairwise Pearson correlation analysis shows that Glucose having strongest correlation with the Outcome variable (r = 0.49), followed by BMI (r = 0.31), Age (r = 0.24), and Pregnancies (r = 0.22). An inter-feature correlation (r= 0.54) exists between Age and Pregnancies .

4.1.3 Model Performance Comparison

All three models were evaluated on the testing dataset (154 samples). Table 3 summarizes the results.

Table 3: Model Performance Comparison (%) — Random Forest is the best performing model

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	70.78	60.00	50.00	54.55
Random Forest	75.32	67.39	57.41	62.00
SVM	74.03	65.22	55.56	60.00

Random Forest achieved the best performance across all four metrics. SVM performed second best on accuracy (74.03%) and recall (55.56% vs. 57.41%) and F1-score (60.00% vs. 62.00%). Overall, logistic regression had the worst performance of all the three algorithms with 70.78% accuracy and only 50.00% recall.

4.1.4 Confusion Matrix Analysis

Logistic regression gave the highest number of false negatives(27). This makes it least suitable for clinical screening. Random forest worked the best by correctly identifying 85 out of 100 non-diabetic patients and 31 out of 54 diabetic patients. SVM correctly identified 30 diabetic patients but wrongly classified 24 patients.

4.1.5 ROC Curve and AUC Analysis

Both Logistic Regression and Random Forest acheived an Area under the curve of 0.813, while SVM scored 0.796. All models outperformed a random classifier (AUC = 0.5).

4.1.6 Feature Importance — Random Forest

According to the feature importance score, Glucose was the most significant predictor (∼29%), followed by BMI (∼16%), DiabetesPedigreeFunction (∼12%), Age (∼11.5%), and Insulin (∼9.5%). SkinThickness and BloodPressure contribute as the least significant to prediction.

4.2 Discussion

4.2.1 *Why Random Forest Performed Best*

Random Forest performed best among both Logistic Regression and SVM across all evaluation metrics. About 100 decision trees trained on random feature subsets, Random Forest is inherently resistant to overfiitting—a critical advantage given the small dataset size (768 samples) [3]. Additionally, Random Forest captures non-linear relationships between features, whereas Logistic Regression assumes linearity and SVM with the RBF kernel, while capable of nonlinear mapping, is more sensitive to hyperparameter selection. This finding is consistent with Afolabi et al. [5], who also reported that tree-based models reliably handled non-linear patient health data.

The worst performance was by Random forest with the highest false negative count (27), mostly due to its linear decision boundary being insufficient for the complex feature interactions present in clinical data. SVM showed good accuracy but was outperformed on recall and F1score.

4.2.2 *Recall as the Critical Metric*

In clinical diabetes screening, recall is the most important metric because a false negative , i.e. classifying a diabetic patient as non-diabetic delays diagnosis and treatment, leading to serious health consequences. The best recall was achieved by Random forest at 57.41%, correctly identifying 31 out of 54 diabetic patients. This is the best among the three models, the suboptimal recall across all models (50%–57%) shows the need for further improvement through class imbalance handling techniques, as demonstrated by Banday et al. [8] and Rahman et al.
[9].

4.2.3 *Impact of Class Imbalance*

All the models were performing much better on majority class after the 65:35 class imbalance.85% of non diabetic patients and only 57% of diabetic patients were classified by random Forest correctly.This disparity occurs when standard classification algorithms are applied to imbalanced datasets without corrective measures. Ganie et al. [10] demonstrated that SMOTE-based balancing can significantly improve boosting model performance on the same dataset.

4.2.4 *Glucose as the Dominant Feature*

Feature importance analysis confirms Glucose as the most influential predictor . It is contributing approximately 29% to the Random Forest model's decisions. This aligns with clinical knowledge, as blood glucose level is the primary diagnostic criterion for diabetes. BMI and DiabetesPedigreeFunction, represent metabolic and genetic risk factors respectively, thus are the next most important features. SkinThickness and BloodPressure hav low importance,

which may be attributed to the high rate of zero values in these columns and their weaker correlation with the outcome.

4.2.5 AUC Interpretation

Logistic Regression and Random Forest share identical AUC values (0.813), showing equivalent ranking ability across all classification thresholds, but, at the default threshold of 0.5, Random Forest demonstrates superior precision, recall, and F1-score. This shows that Random Forest makes better calibrated predictions at standard operating point.

5 Limitations

This study has several limitations that should be acknowledged:

(i) Sub-optimal Recall: The recall values achieved by all the three models values between 50-57%, thus significant portion of diabetic patients are missed. This can likely be due on not applying class imbalance techniques.

(ii) Dataset Bias: The Pima Indians Diabetes dataset exclusively contains records of female patients of aged 21 and above, limmiting generalizability to other demographic groups, as also noted by Banday et al. [8].

(iii) Small Dataset Size: The dataset is relativeling small with only 768 sampled. This may cause high varianceing in the model estimates.

(iv) No Hyperpara meter Optimization: Traininging of models was done with default or manually selected hyperparametersing. Systematic tuning using grid search or Bayesian optimizationed was not performed, unlike the approach taken by Jain & Singhal [13].

(v) Limited Feature Set: The dataset containing only 8 clinical features. More fea;;tures such as HbA1c, fasting glucoseing a, or lifestyle factors could improve predictive performance of models, as suggested by Wee et al. [7].

(vi) Black-box Limitation Not Addressed: While this study uses interpre ;table classical models, the feature importance analysis is limiiteding; to Random Forest's built-in method. More advanced explaii nability techniques such as SHAP or LIME were not employed, as highlighted by Adlakha et al. [14].

6 Conclusion and Future Scope

6.1 Conclusion

In this study, we did a syste matic comparative eval ;uation of three supervised machine learning models namely Logistic regression, Random forest, and Support Vector Machine for

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The following directions are recommended for future work:

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Declarations

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Conflict of Interest: The authors declare no conflict of interest .

Data Availability: The Pima Indians Diabetes Dataset is publicly available from the UCI Machine Learning Repository and Kaggle.

Code Availability: The complete implementation code is available as a Jupyter Notebook submitted along with this report.

Author Contribution: All authors contributed equally to this work.

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